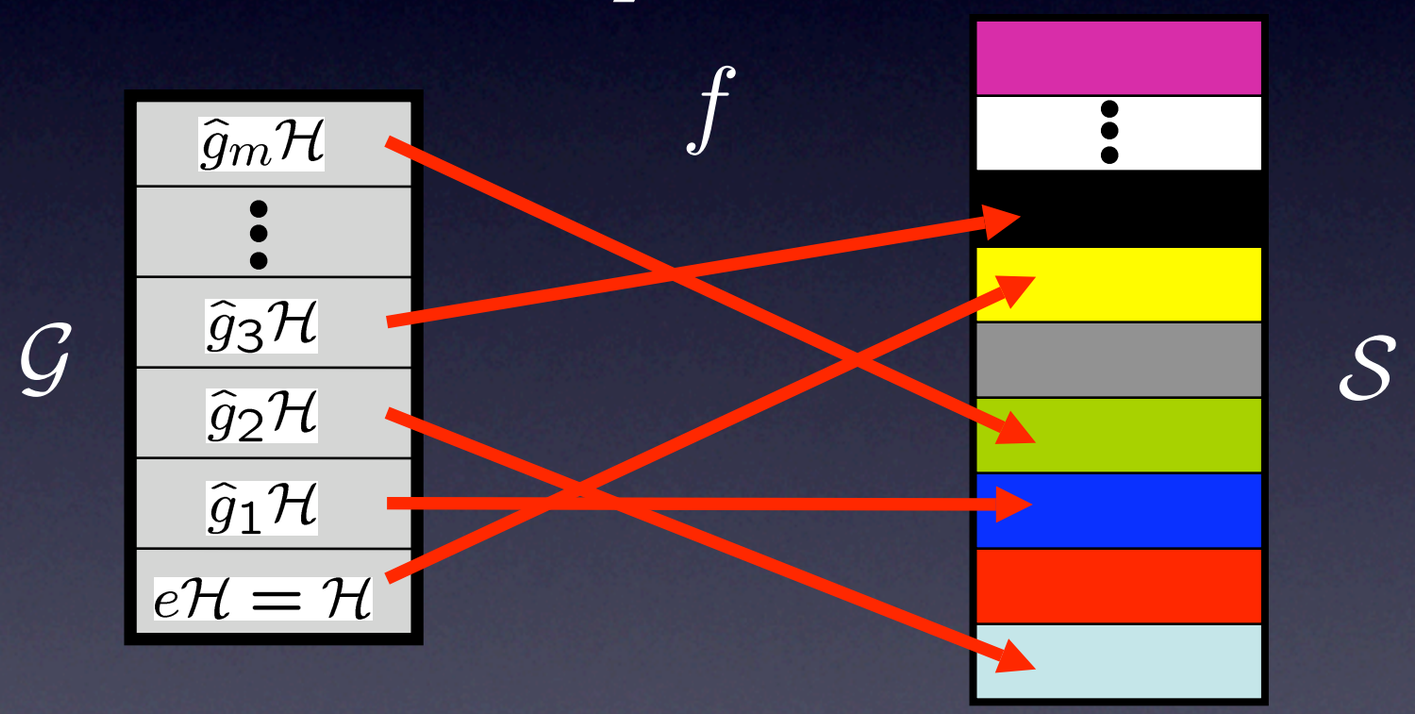
The Hidden Subgroup Conjugacy Problem and Beyond



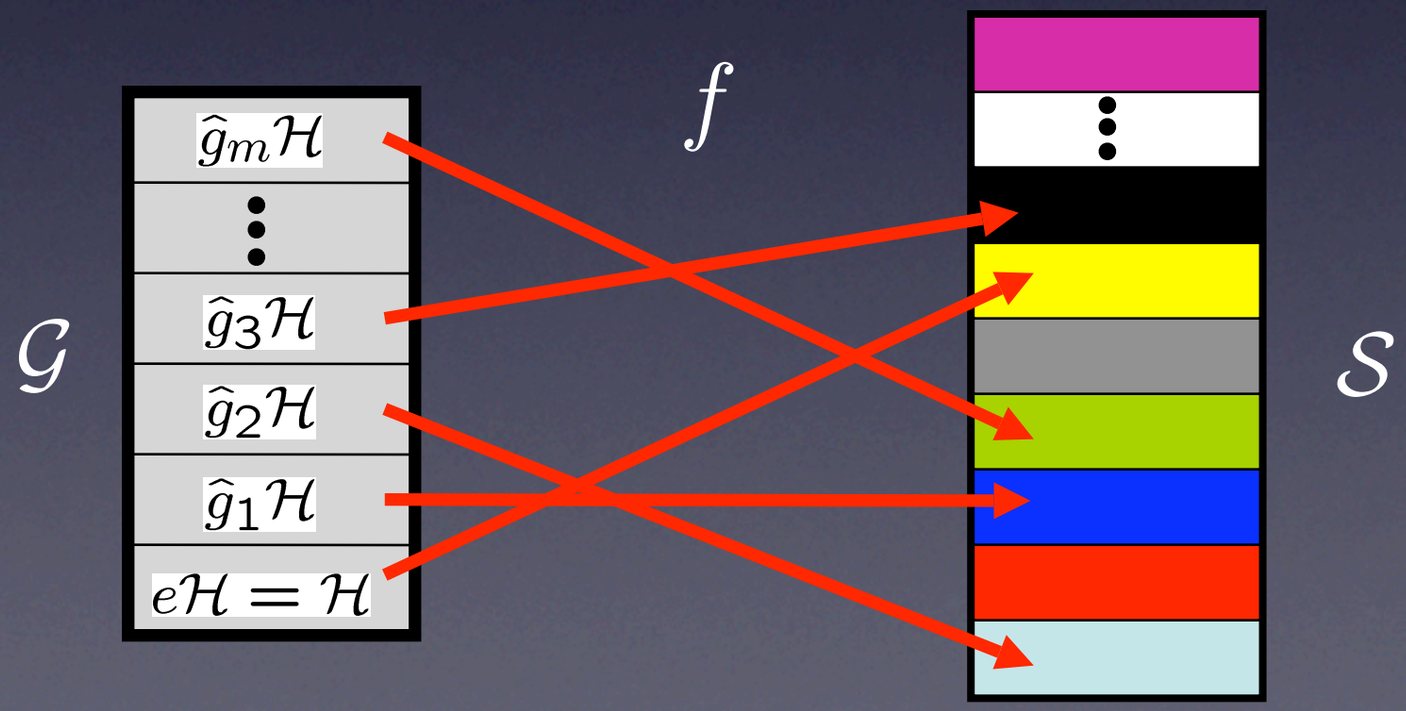
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The Hidden Subgroup Problem

Given: a function f, from a group $\mathcal{G}$, to a set $\mathcal{S}$, which is constant and distinct on left cosets of a hidden subgroup $\mathcal{H}$.

Problem: identify $\mathcal{H}$ by querying f

Problem size: $\log |\mathcal{G}|$



Reminder

- Shor's algorithm uses an efficient algorithm for the Abelian HSP
- Efficient algorithm for HSP over S_n would yield an efficient algorithm for graph isomorphism
- Efficient algorithm for HSP over dihedral group would yield an efficient algorithm for certain unique shortest vector in a lattice problems (Regev)

The Hidden Subgroup Conjugacy Problem

Given: a function f, from a group $\mathcal{G}$, to a set $\mathcal{S}$, which is constant and distinct on left cosets of a hidden subgroup $\mathcal{H}$.

Problem: identify $[\mathcal{H}]$ by querying f

$$[\mathcal{H}] = \{\mathcal{J} | \exists g \in G, \mathcal{J} = \mathcal{H}^g\}$$

$$\mathcal{H}^g = \{ghg^{-1} | h \in \mathcal{H}\}$$

Problem size: $\log |\mathcal{G}|$

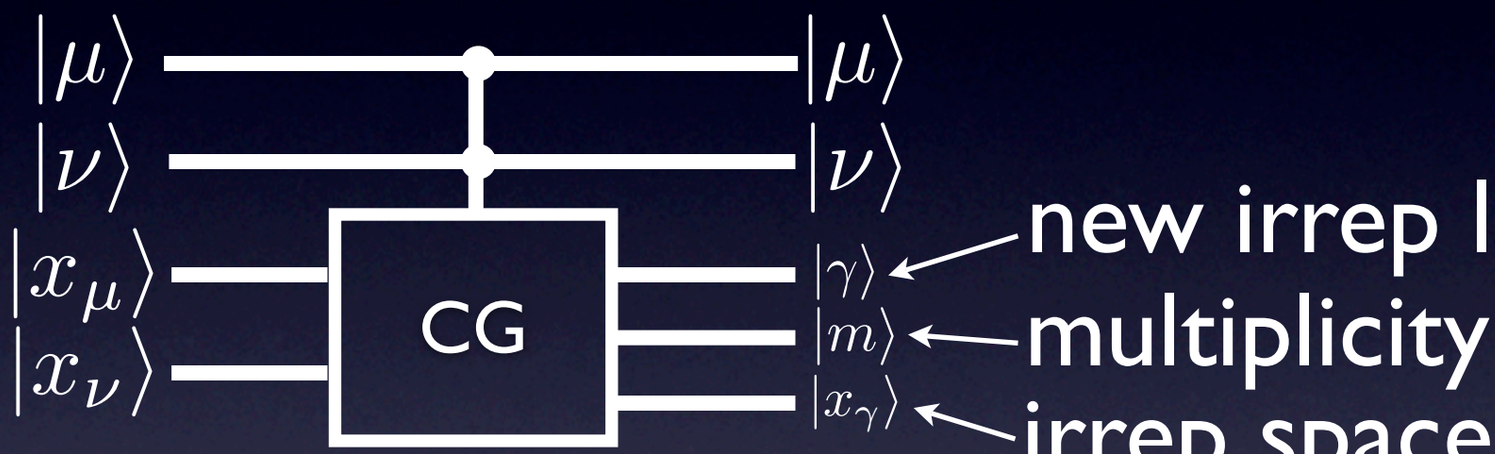
Symmetries

Hidden subgroup state	Symmetry	Transform
$\rho_{\mathcal{H}} = \frac{\lvert\mathcal{H}\rvert}{\lvert\mathcal{G}\rvert} \sum_{\hat{g}} \lvert\hat{g}\mathcal{H}\rangle\langle\hat{g}\mathcal{H}\rvert$	$D_L(g)\rho_{\mathcal{H}} = \rho_{\mathcal{H}} D_L(g)$	QFT over $\mathcal{G}$

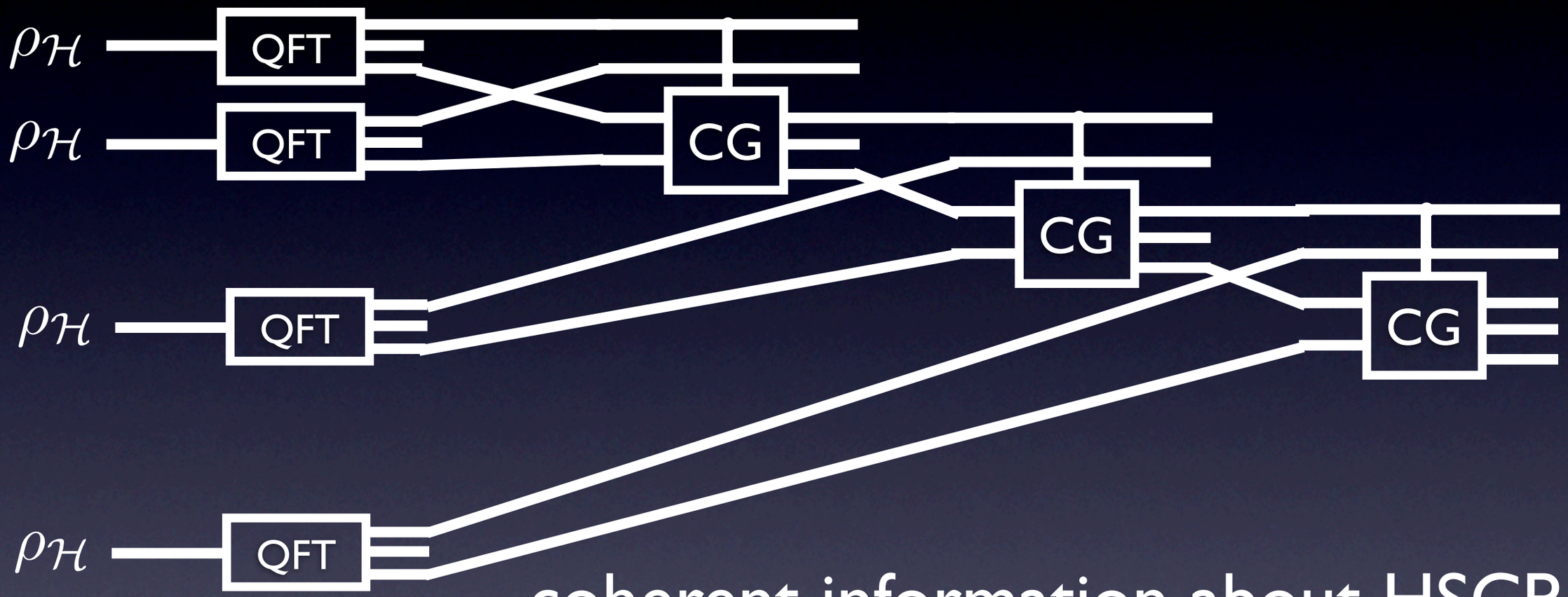
Hidden subgroup conjugacy states	Symmetry	Transform
$\rho_{[\mathcal{H}]} = \frac{1}{\lvert\mathcal{G}\rvert} \sum_{g\in\mathcal{G}} \bigotimes_{i=1}^{n} \rho_{\mathcal{H}^g}$	$\rho_{[\mathcal{H}]} D_R(g)^{\otimes n} = D_R(g)^{\otimes n} \rho_{[\mathcal{H}]}$	Clebsch-Gordan transform over $\mathcal{G}$

Clebsch-Gordan Transform

$$D_\mu(g) \otimes D_\nu(g) \cong \bigoplus_\gamma I_{n^\gamma_{\mu,\nu}} \otimes D_\gamma(g)$$



Clebsch-Gordan Symmetry



coherent information about HSCP

output: $|\mu_1, \mu_2, \ldots, \mu_n\rangle|\mu'\rangle|m'_2, m'_3 \ldots m'_n\rangle|x'\rangle$

classical information no HSCP info

[Bacon Quantum Inf. and Comp., 8, 438-467 (2008)]

Simon's Algorithm Symmetry

Symmetry ⇒ apply Hadamard transform

$$|0\rangle \qquad |t\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \qquad \text{trivial irrep}$$

$$\Rightarrow$$

$$|1\rangle \qquad |a\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) \qquad \text{alternating irrep}$$

Abelian group CG: $D_\mu(g) \otimes D_\nu(g) = D_{\mu+\nu}(g)$

CG $\mathbb{Z}_2$:

$$\begin{aligned} D_t(g) \otimes D_t(g) &= D_t(g) \\ D_t(g) \otimes D_a(g) &= D_a(g) \\ D_a(g) \otimes D_t(g) &= D_a(g) \\ D_a(g) \otimes D_a(g) &= D_t(g) \end{aligned}$$

Gaussian elimination stage is nothing more than a Clebsch-Gordan seive!

Generalized Simon's Problem

arXiv:quant-ph/0603251:
"Quantum Algorithms for Simon's Problem Over General Groups" by Alagic, Moore, and Russell

Subexponential time algorithms for finding hidden involution ($g^2=I$) over $\mathcal{G}^n$.

Uses Clebsch-Gordan sieving on irreps.

But we know groups for which $\mathcal{G}^n$ admits efficient quantum algortihms. Do CG sieves work for them?

Hidden Symmetries of Multiple Squares



Hidden Subgroup Problem for $\mathcal{D}_4{}^n$ with subgroups of order 2 which contain a reflection

$\mathcal{D}_4{}^n$ is smoothly solvable (derived series is of constant length and bounded exponent) and so efficient algo in:

arXiv:quant-ph/0211091
"Hidden Translation and Orbit Coset in Quantum Computing" Friedl, Ivanyos, Magniez, Santha, and Sen

Symmetries of a Square

$$\mathcal{D}_4 = \langle r, s | r^2 = e, s^4 = e, rsrs = e \rangle$$

$$r^t s^k, t \in \{0, 1\}, k \in \{0, 1, 2, 3\}$$

representations:

$$D_j(r^t s^k) = \begin{bmatrix} \omega^{jk}\delta_{t,0} & \omega^{-jk}\delta_{t,1} \\ \omega^{jk}\delta_{t,1} & \omega^{-jk}\delta_{t,0} \end{bmatrix} \quad j \in \{0, 1, 2, 3\}$$

reducible: $D_0 = D_t \oplus D_a$

Clebsch-Gordan series:

$$D_i \otimes D_j = D_{i+j} \oplus D_{i-j}$$

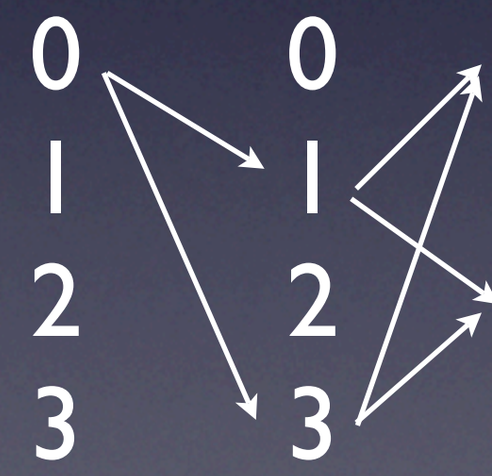
Symmetries of a Square

Hidden subgroup state (hides l): $|\psi_{i,l}\rangle = \frac{1}{\sqrt{2}}(|0\rangle + \omega^{il}|1\rangle)$

Irrep 2 can be used to find parity information about l

$$|\psi_{i,2}\rangle = \frac{1}{\sqrt{2}}(|0\rangle + (-1)^l|1\rangle)$$

$|\psi_{i,0}\rangle \otimes |\psi_{i,1}\rangle \otimes |\psi_{i,1}\rangle$



We can use Clebsch-Gordan transform to distill out 0 or 2 irreps from non 0/2 irreps

Parity information solve HSCP. To solve HSP compute extra function when querying HSP.

Symmetries of a Squares

Must run same algorithm in parallel

hidden involution: $\times_{i=1}^{n} r^{t_i} s^{l_i}$

$$(t_i, l_i) \in \{(0,0), (1,0), (1,1), (1,2), (1,3)\}$$

Three steps

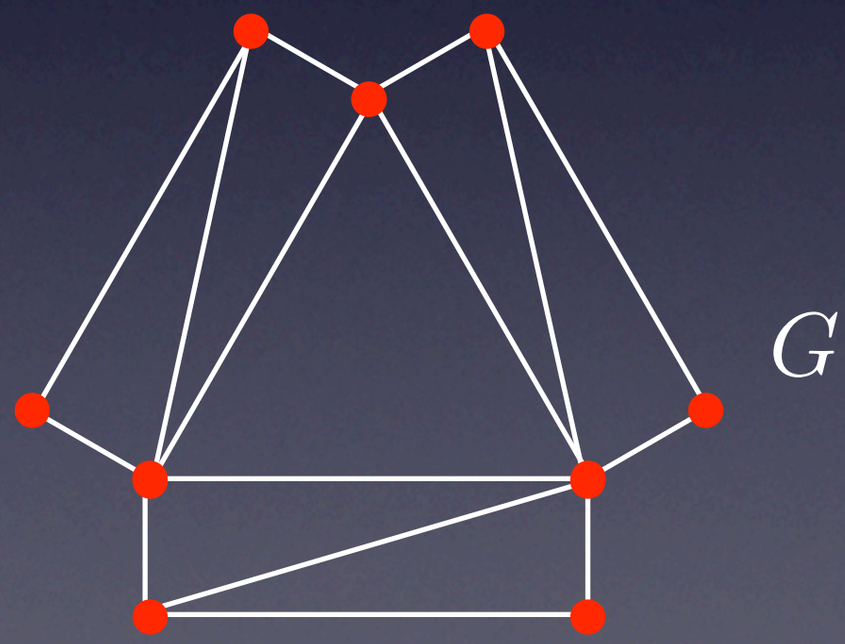
1. Determine $t \in \{0,1\}^n$
2. Determine parity of $l \in \{0,1,2,3\}^n$
3. Determine $l \in \{0,1,2,3\}^n$

All three steps use Clebsch-Gordan transforms

Graph Automorphism

Given: a graph G

Problem: Does there exist a nontrivial subgroup of the symmetric group acting on the vertices of G which preserves the graph's edge structure?



G

(at least as hard as graph isomorphism)

S_n HSP and Graph Automorphism

Hidden subgroup problem on symmetric group

function: $f(\pi) = A^\pi$ A adjacency matrix of graph

$$A^\pi = P(\pi)AP(\pi^{-1})$$

Problem: decide if hidden subgroup is non-trivial

HSP and Graph Automorphism

There are now numerous efficient quantum algorithms for non-Abelian HSPs.

HSPs which admit efficient quantum algorithms:

- almost abelian groups
- near Hamiltonian groups
- normal subgroups
- small commutator subgroup
- solvable groups of bounded exponent and of bounded derived series
- groups of nilpotency class at most 2 (including extraspecial groups)
- groups with an Elementary Abelian Normal 2-Subgroup of Small Index

Are any of these useful for Graph Automorphism?

work with Will Johnson (UW)

HSP and Graph Automorphism

Some groups have hidden subgroup problem which have widely resisted efficient quantum algorithms.

- Dihedral group
- Certain Metacyclic groups

Are any of these useful for Graph Automorphism?

Dihedral Group into S_n

Efficient permutation representation of some dihedral groups

$$s = (123)(45678)\cdots(\sum_{i=1}^{k-1} p_i + 1 \cdots \sum_{i=1}^{k} p_i)$$

$$r = (13)(48)(57)\cdots(\sum_{i=1}^{k-1} p_i + 1, \sum_{i=1}^{k} p_i)$$

Represented as permutation representation in $S_{\sum_{i=1}^{k} p_i}$ but size of group is $2\prod_{i=1}^{k} p_i$



Dihedral HSP and GA

Suppose we had efficient algorithm for HSP over dihedral group, could this be used to solve (restricted) graph automorphism problems?

Given: Graph G and action of dihedral group on G specified by two generators.

Problem: Find $Aut(G) \cap \mathcal{D}'$

Dihedral HSP and GA

$G = (V, E)$

1. Partition V into orbits Y_i
2. For each Y_i, Y_j (including $i = j$)
   1. Find subgroup of local group which acts as automorphism
   2. Lift to original dihedral group and add to running list of subgroups
3. Take intersection of list of subgroups

When efficient embedding into permutation group exists the above classical algorithm solves this restricted GA

Normal Subgroups?

There exist efficient quantum algorithms for solving HSP when subgroups are normal. Useful?

If $\mathcal{H} = Aut(G) \cap \mathcal{G} \lhd \mathcal{G}$ then there is an efficient classical algorithm which finds $\mathcal{H}$

Normal subgroup HS algorithm not useful for GA

Plethora Not Useful

Given: Graph $G = (V, E)$ and permutation group $\mathcal{G} \subset \mathrm{Sym}(V)$

Use efficient quantum algorithm over $\mathcal{G}$ to find $\mathcal{G} \cap Aut(G)$?

HSPs which admit efficient quantum algorithms are all not useful for HSP:

- almost abelian groups
- near Hamiltonian groups
- normal subgroups
- small commutator subgroup
- solvable groups of bounded exponent and of bounded derived series
- groups of nilpotency class at most 2 (including extraspecial groups)
- groups with an Elementary Abelian Normal 2-Subgroup of Small Index

- Hidden symmetries of multiple squares (arXiv:0808.0174)
- Current efficient non-Abelian hidden subgroup problem algorithms are not useful for Graph Automorphism? (with W. Johnson, in preparation)

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